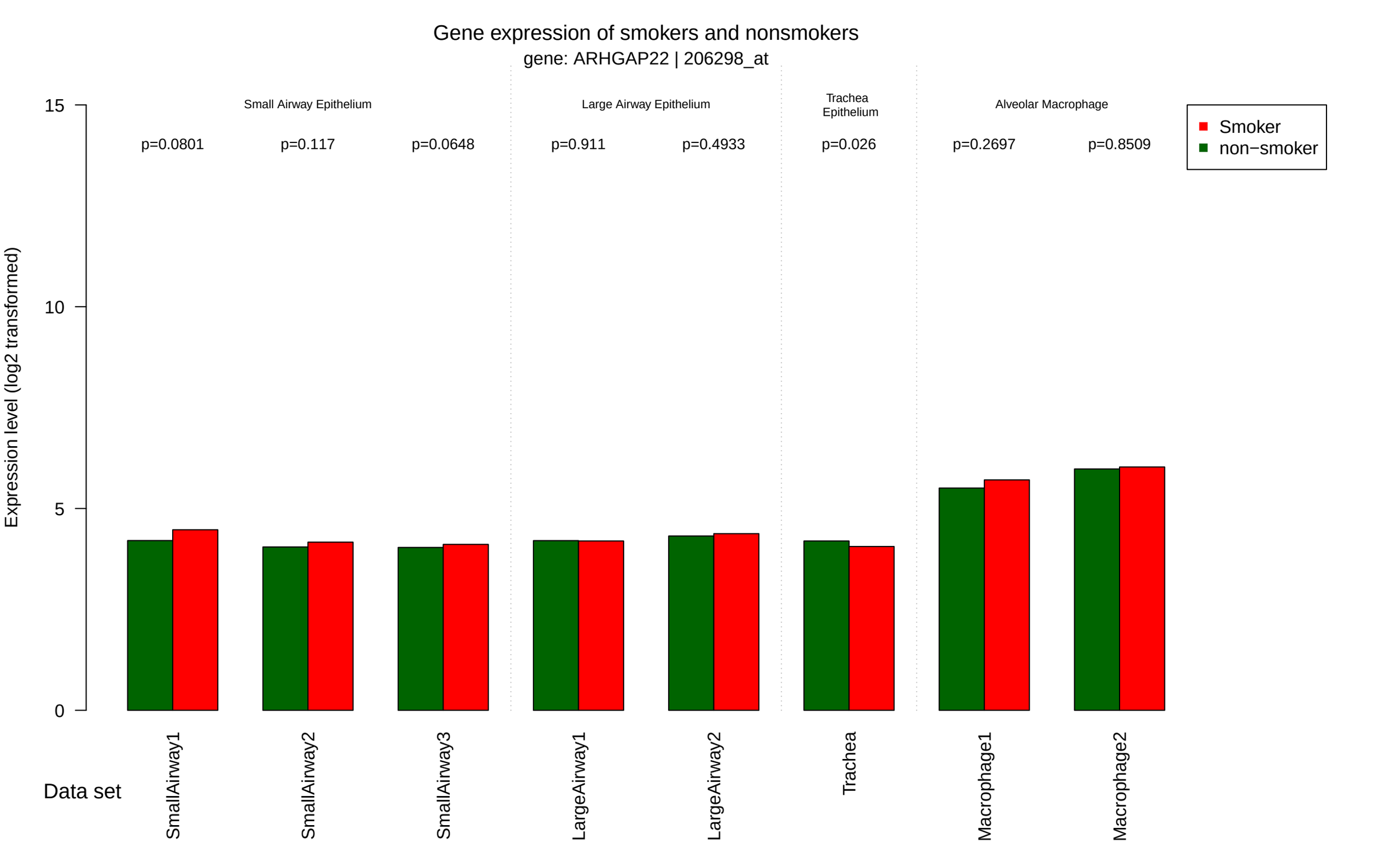
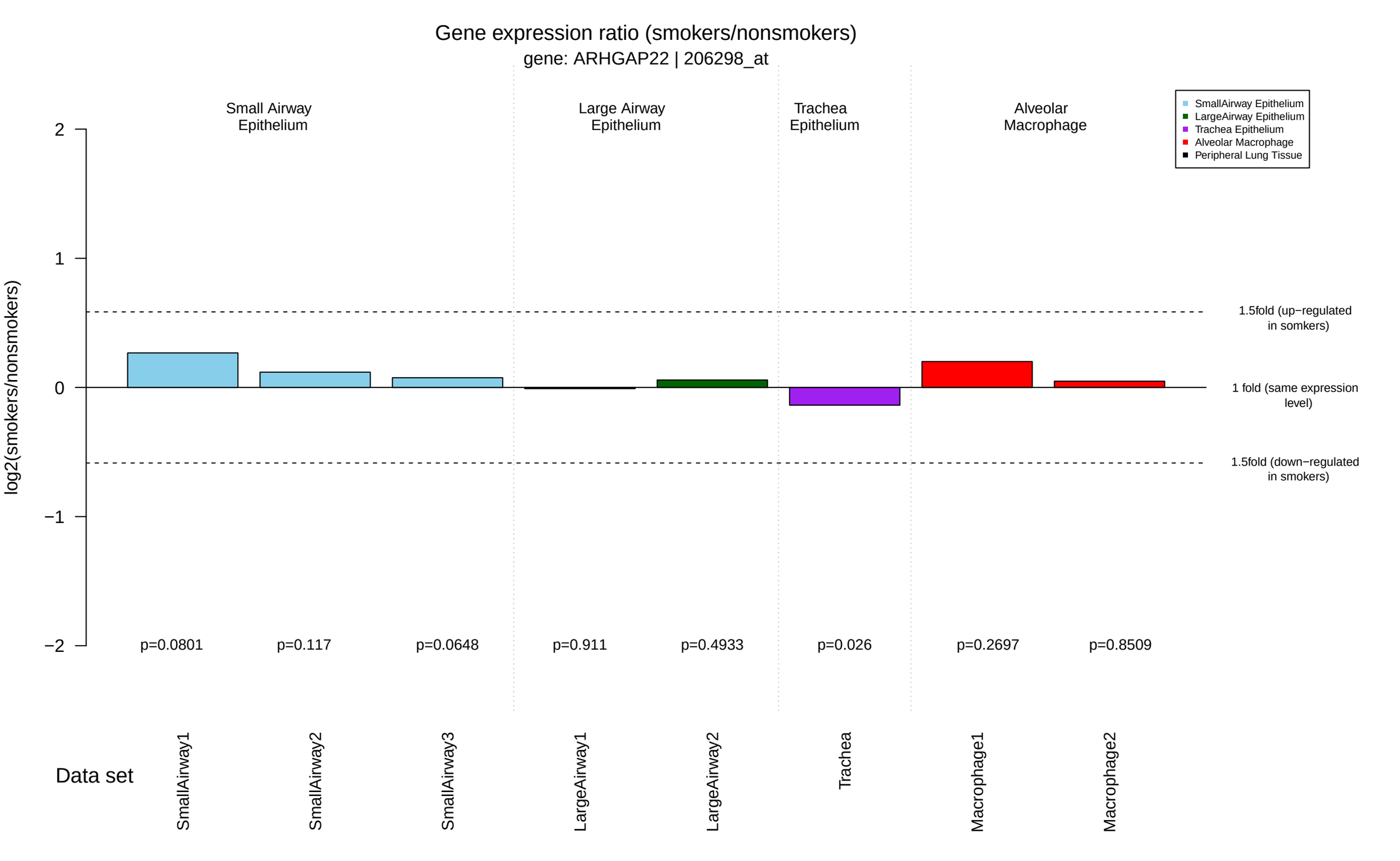
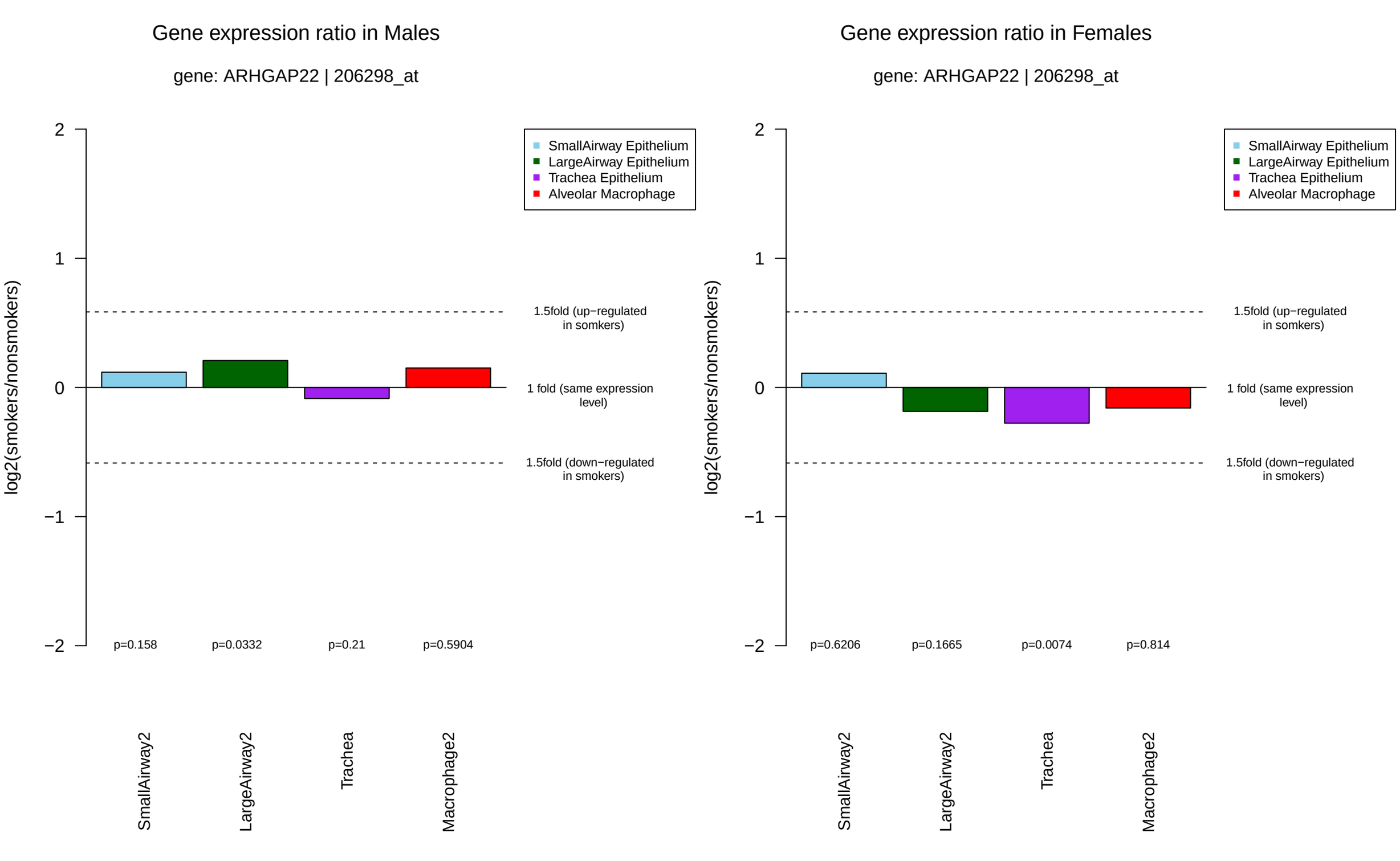
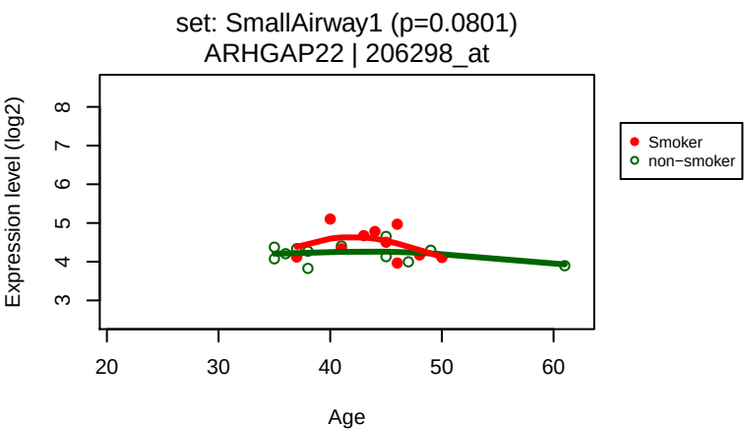
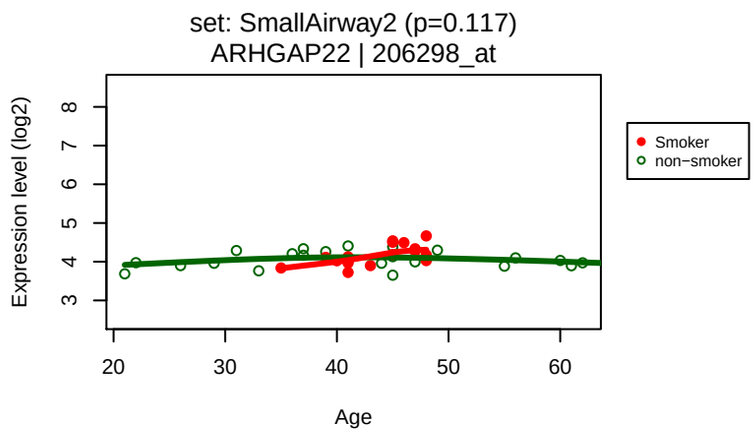
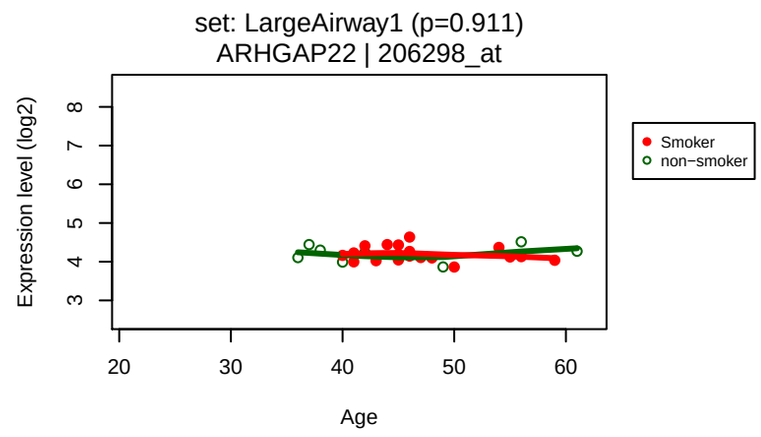

Gene expression of smokers and nonsmokers
gene: ARHGAP22 | 206298_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0801
p=0.117
p=0.0648
p=0.911
p=0.4933
p=0.026
p=0.2697
p=0.8509
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

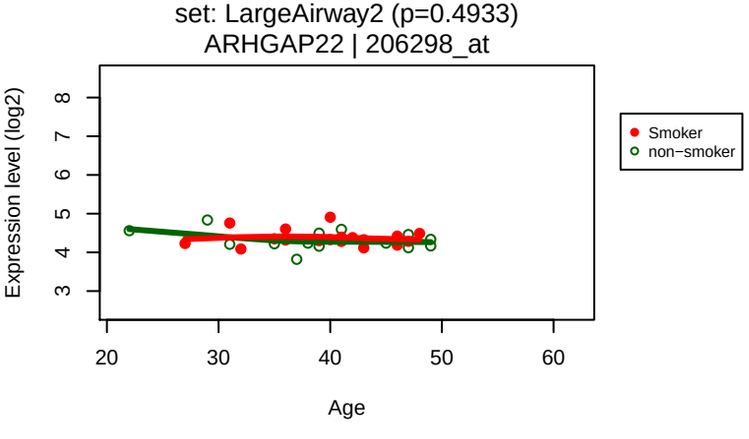

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP22 | 206298_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0801
p=0.117
p=0.0648
p=0.911
p=0.4933
p=0.026
p=0.2697
p=0.8509
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

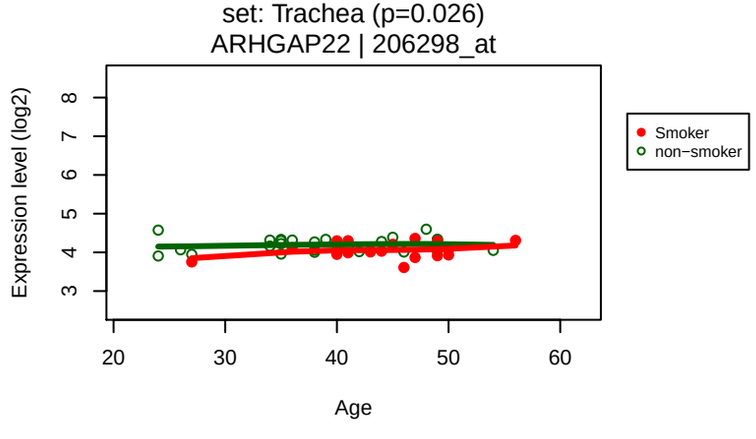

Gene expression ratio in Males
gene: ARHGAP22 | 206298_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.158
p=0.0332
p=0.21
p=0.5904
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP22 | 206298_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6206
p=0.1665
p=0.0074
p=0.814
SmallAirway2
LargeAirway2
Trachea
Macrophage2

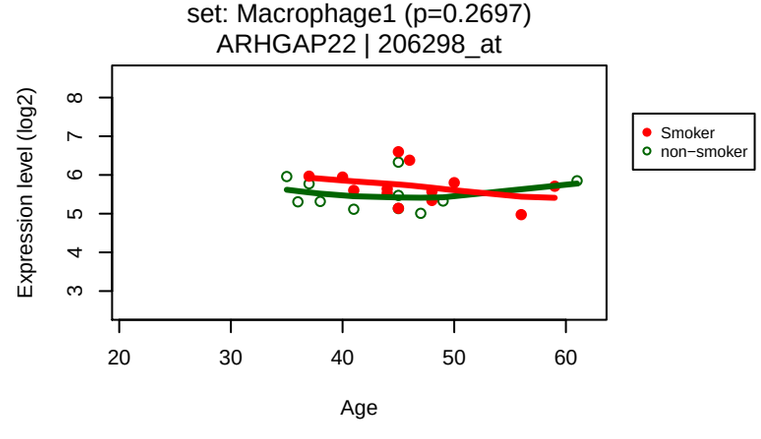

set: SmallAirway1 (p=0.0801)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker

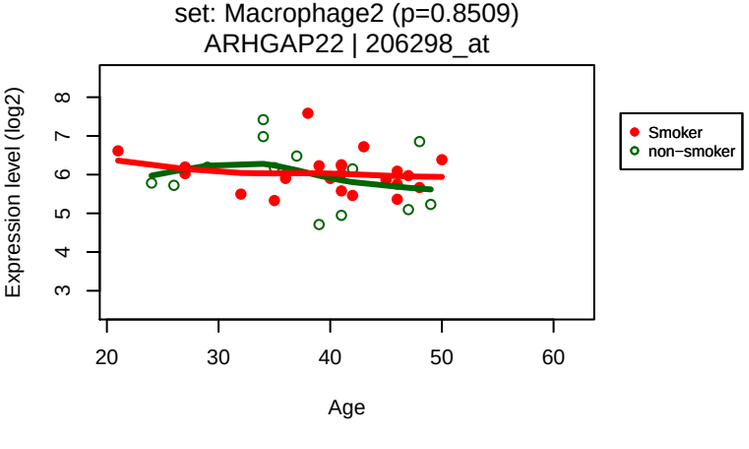

set: SmallAirway2 (p=0.117)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker

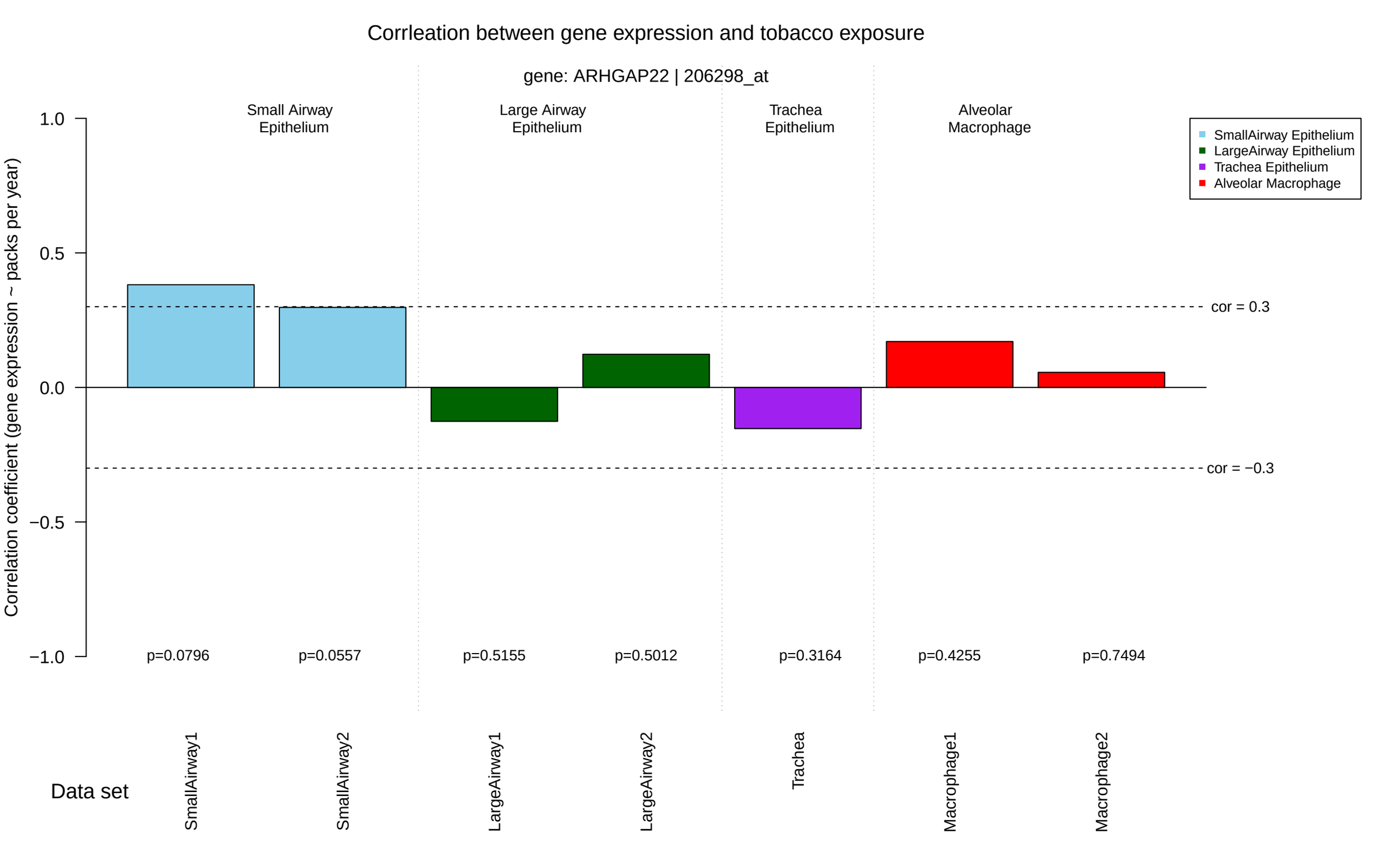

set: LargeAirway1 (p=0.911)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4933)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.026)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2697)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8509)
ARHGAP22 | 206298_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP22 | 206298_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0796
p=0.0557
p=0.5155
p=0.5012
p=0.3164
p=0.4255
p=0.7494
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2